Name _____ Period _____ Date _____

The Puppy Puzzle

Focus question: How can puppies from the same parents be alike, but not identical?

Warm-up

What do you notice about the puppies and their parents? What do you wonder?

Notice and wonder

I notice ... I wonder ...

Compare the traits

Glue in: Puppy litter data

My first model

Draw Parent 1, Parent 2, and two offspring. Use arrows to show information moving from the parents to the offspring. Label one shared trait and one different trait.

Name _____ Period _____ Date _____

The Puppy Puzzle (continued)

Focus question: How can puppies from the same parents be alike, but not identical?

Three possible explanations

Three small sketch boxes: one idea in each (information mixed, information changes, environment has an effect).

Key idea

Parents -> ____ -> Offspring -> similarities and ____

Exit ticket

One explanation for why siblings can be different is ____. The evidence from the puppies is ____.

Name _____ Period _____ Date _____

Mixing Information

Focus question: If offspring get information from the same two parents, why might each get a different combination?

Warm-up

Idea 1: information is ____ in different ways. Idea 2: information ____. Idea 3: the ____ has an effect.

Make possible offspring

Glue in: Card table: 3 offspring

Analyze the pattern

Every offspring got information from ____ parents. Each offspring got a ____ combination of cards. Different combinations can lead to different ____.

Name _____ Period _____ Date _____

Mixing Information (continued)

Focus question: If offspring get information from the same two parents, why might each get a different combination?

Revise our model

Revise your Day 1 model: Parent 1, Parent 2, information from each parent, the information being combined, two possible offspring, one similarity, one difference.

Three sources of variation

1. New ____ of alleles. 2. ____ when DNA is copied. 3. Mutations caused by the ____.

Exit ticket

Name the three sources of new genetic variation.

Name _____ Period _____ Date _____

Mixing Information: Reading Notes

Focus question: What does the reading say about genes and variation?

Warm-up

Skim the headings. What do you predict the reading is about?

Reading: Lesson 13.1 Genes and Variation (pp. 370–374)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (allele, variation, gene pool)

Name _____ Period _____ Date _____

Mixing Information: Reading Notes (continued)

Focus question: What does the reading say about genes and variation?

Main idea in my words

Question I still have

Connect to the puppies

Sketch one way the reading explains why puppies from the same parents differ.

Exit ticket

One new idea from the reading is ____.

Name _____ Period _____ Date _____

Genetics Notes, Part 1: Genes and Alleles

Focus question: What are genes and alleles, and what do dominant and recessive mean?

Warm-up

A ____ is one long DNA molecule. A ____ is a section of DNA that codes for one protein, and proteins help produce a ____.

Heredity

Chromosomes

Genes make proteins make traits

Alleles

Sketch a pair of chromosomes. Color each set of alleles a different color.

Name _____ Period _____ Date _____

Genetics Notes, Part 1: Genes and Alleles (continued)

Focus question: What are genes and alleles, and what do dominant and recessive mean?

Dominant vs. recessive

Black or brown cats

Draw three cat heads: BB, Bb, bb. Color each one to show its coat.

Exit ticket

An allele that shows up when only one copy is present is ____. A recessive trait shows only when ____ copies are present.

Name _____ Period _____ Date _____

Genetics Notes, Part 1: Genes and Alleles: Reading Notes

Focus question: How did Mendel figure out dominant and recessive traits?

Warm-up

What do you already know about Gregor Mendel?

Reading: Lesson 10.1 The Work of Gregor Mendel (pp. 286–290)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (heredity, allele, dominant, recessive)

Name _____ Period _____ Date _____

Genetics Notes, Part 1: Genes and Alleles: Reading Notes (continued)

Focus question: How did Mendel figure out dominant and recessive traits?

Main idea in my words

Question I still have

Connect to the cats

Sketch how Mendel's results could explain a black cat with a brown parent.

Exit ticket

One thing Mendel showed is ____.

Name _____ Period _____ Date _____

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares

Focus question: How can a Punnett square show the possible allele combinations?

Warm-up

A dominant allele is written with a ____ letter. A recessive allele is written with a ____ letter.

Genotype and phenotype

Glue in: Genotype to phenotype: pea plants. Draw two eyes: color one blue and one brown.

Homozygous and heterozygous

Write CC, Ee, Gg, dd, AA, Hh. Color the homozygous ones green and the heterozygous ones orange. Then write the missing words.

Name _____ Period _____ Date _____

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares (continued)

Focus question: How can a Punnett square show the possible allele combinations?

Punnett squares

Glue in: Punnett squares: steps and worked example

Practice

Cross 1: Tt × tt. Cross 2: TT × Tt. ____ of the 4 boxes are tall and ____ are short.

Exit ticket

In a Tt × Tt cross, ____ out of 4 boxes show a short plant (tt).

Name _____ Period _____ Date _____

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares: Reading Notes

Focus question: How do Mendel's principles apply to many traits?

Warm-up

What is the difference between genotype and phenotype? Guess.

Reading: Lesson 10.2 Applying Mendel's Principles (pp. 291–296)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (genotype, phenotype, homozygous, heterozygous, probability)

Name _____ Period _____ Date _____

Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares: Reading Notes (continued)

Focus question: How do Mendel's principles apply to many traits?

Main idea in my words

Question I still have

Try it

Use a Punnett square from the reading to check one result.

Exit ticket

One way Punnett squares help is ____.

Name _____ Period _____ Date _____

Monster Genetics, Part 1: Flip a Coin, Build a Monster

Focus question: How can two coin flips build a monster's genotype?

Warm-up

Each sex cell gets ____ allele. A coin flip can choose between B and ____.

How the lab works

Glue in: Monster Genetics: the coin rule and Table 2

Build the female monster

Use Table 1 on your Monster Genetics sheet.

Name _____ Period _____ Date _____

Monster Genetics, Part 1: Flip a Coin, Build a Monster (continued)

Focus question: How can two coin flips build a monster's genotype?

Fill in the male monster

Use Table 2 (glued in at left) and your Monster Genetics sheet.

Compare

Our monsters are different because the flips gave different ____ of alleles.

Exit ticket

Each coin flip stood for the ____ a sex cell passes on. Two flips gave the monster's ____.

Name _____ Period _____ Date _____

Monster Genetics, Part 2: Predict the Baby

Focus question: How do sex cells and fertilization mix alleles, and how can we predict the baby?

Warm-up

A Gg parent makes sex cells with ____ or ____.

Sex cells and fertilization

Meiosis makes ____ with one allele of each gene. Crossing over happens when paired chromosomes ____ pieces of DNA. Fertilization joins two sex cells to make a new ____ of alleles.

Punnett squares for the baby

Glue in: Punnett square: skin color and eyes

Name _____ Period _____ Date _____

Monster Genetics, Part 2: Predict the Baby (continued)

Focus question: How do sex cells and fertilization mix alleles, and how can we predict the baby?

Percent chance

In a Gg × Gg cross, ____% of the boxes are green and ____% are blue.

Class data

Glue in: Prediction vs. class count

Exit ticket

Describe how sex cells and fertilization make new allele combinations.

Name _____ Period _____ Date _____

Monster Genetics, Part 2: Predict the Baby: Reading Notes

Focus question: How does meiosis make sex cells?

Warm-up

How many alleles of a gene are in an egg cell? Guess.

Reading: Lesson 10.3 Meiosis (pp. 301–307)

Biology: The Living Earth. Skip the phases of meiosis. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (meiosis, gamete, crossing over, independent assortment)

Name _____ Period _____ Date _____

Monster Genetics, Part 2: Predict the Baby: Reading Notes (continued)

Focus question: How does meiosis make sex cells?

Main idea in my words

Question I still have

Sketch it

Sketch a pair of chromosomes swapping one piece (crossing over).

Exit ticket

Meiosis mixes alleles because ____.

Name _____ Period _____ Date _____

When Copying Goes Wrong

Focus question: How do copying errors and UV light change genes, and when are the changes inherited?

Warm-up

A mutation is a ____ in the DNA sequence. A mutation can change an ____ in a protein.

How do mutations happen?

A ____ error that is not fixed becomes a mutation. UV light and some ____ can also damage DNA.

Four kinds of mutations

Glue in: Four kinds of mutations

UV and mutations

Glue in: UV light and mutations

Name _____ Period _____ Date _____

When Copying Goes Wrong (continued)

Focus question: How do copying errors and UV light change genes, and when are the changes inherited?

Who passes it on?

Glue in: Who passes it on? Five cases

Explain it

UV light can change a gene by ____ its DNA. The change is inherited only if it happens in a cell that makes ____.

Exit ticket

Explain one way a mutation can happen and one reason a mutation might not be inherited.

Name _____ Period _____ Date _____

When Copying Goes Wrong: Reading Notes

Focus question: How is DNA copied, and what can go wrong?

Warm-up

What is DNA replication? Guess.

Reading: Lesson 11.3 DNA Replication (pp. 328–331) and Lesson 12.4 Mutations (pp. 357–361)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (replication, mutation, proofreading, mutagen)

Name _____ Period _____ Date _____

When Copying Goes Wrong: Reading Notes (continued)

Focus question: How is DNA copied, and what can go wrong?

Main idea in my words

Question I still have

Connect to the puppies

Sketch where a copying error could enter the puppy story.

Exit ticket

Cells proofread DNA because ____.

Name _____ Period _____ Date _____

Where Did This Variation Come From?

Focus question: How can data show where a genetic variation came from?

Warm-up

Which kind of clue would point to a new combination, a copying error, and an environmental mutation?

Clues for each source

New combination: every allele in the offspring was ____ in a parent. Copying error: a ____ allele that neither parent has. Environmental mutation: ____ mutations in the group that was exposed.

Case 1: the puppy litter

Glue in: Case data: three cases

Case 2: a new base

Use "Case data: three cases". The puppy's CCT allele is ____ because neither parent has it. The most likely source is a ____.

Name _____ Period _____ Date _____

Where Did This Variation Come From? (continued)

Focus question: How can data show where a genetic variation came from?

Case 3: X-rays and mice

Use "Case data: three cases". The high-dose group had ____ mutations. The group with no X-rays had ____. The extra mutations came from ____.

Write your claim

Claim: ... Evidence: ... Reasoning: ...

Exit ticket

Name the source of variation in one case and the evidence.

Name _____ Period _____ Date _____

Defend Your Claim and Back to the Puppies

Focus question: How do I defend my claim against another explanation?

Warm-up

Which three sources of variation have we studied? Name one piece of evidence for each.

How to defend a claim

A counterclaim is ____ explanation. To defend my claim I show why the ____ does not fit the data.

The new coat pattern

Glue in: The new coat pattern: two kennels

Fix my Day 1 model

Redraw your Day 1 model. Add: new combinations, copying errors, mutations from the environment, and which changes are inherited.

Name _____ Period _____ Date _____

Defend Your Claim and Back to the Puppies (continued)

Focus question: How do I defend my claim against another explanation?

Write It: the puppy puzzle

Claim ... Evidence ... Reasoning ...

Make up missing work

Exit ticket

Write one counterclaim and one piece of data that defends against it.

Name _____ Period _____ Date _____

Review Day: Prepare for the CSA

Focus question: Which CFA reviews will best get me ready for the CSA?

Warm-up

List the five unit targets in your own words.

Check my CFAs

Glue in: My target tracker

My reviews

List each review you did and one thing you fixed.

Name _____ Period _____ Date _____

Review Day: Prepare for the CSA (continued)

Focus question: Which CFA reviews will best get me ready for the CSA?

Enrichment: environment and traits

Glue in: Hydrangea soil data

Self-check

Rate each target 1 to 4 on your tracker.

Exit ticket

The target I will study most tonight is ____ because ____.

Name _____ Period _____ Date _____

CSA

Focus question: Can I show I have met the targets for this unit?

Warm-up

Before the CSA, I feel ____ about ____.

What I learned about inherited instructions

After the CSA: three things I understand better now.

The puppy puzzle one more time

Explain the puppy puzzle with the three sources of variation.

Exit ticket

The one idea I want to remember from this unit is ____.